

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

Student Name: Sautrik Roy	Reg No: RA2511003010052
Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit II | Session 5 | SLO 1: Exception Handling (try-catch-finally) (Code: 2051)

Q: Activity: Implement a robust exception handling program illustrating try, multiple catch blocks, and the finally block for resource cleanup.

Ans: Java Implementation:

```
public class ExceptionHandlingDemo {
    public static void main(String[] args) {
        try {
            int[] arr = {10, 20, 30};
            int numerator = 100;
            int denominator = 0;
            // Triggers ArithmeticException
            int result = numerator / denominator;
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            System.err.println("Caught Math Exception: Division by zero!");
        } catch (ArrayIndexOutOfBoundsException e) {
            System.err.println("Caught Array Index Exception!");
        } finally {
            System.out.println("Finally block executed: Cleaned up resources.");
        }
    }
}
```